

Research Proposal

Project Title:

Full-Modality Pretraining and Agent-in-the-Loop
Post-Training for Scalable Vision-Language-Action Models

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1 Introduction

1.1 Background

Large language models (LLMs) and vision-language models (VLMs) have transformed how machines process text, images, and code, yet their impact on the physical world remains limited. The next frontier is to extend these capabilities beyond screens and into physical environments, enabling robots to perceive, reason about, and act upon the real world with the same generality that LLMs exhibit in language. Vision-Language-Action (VLA) models are the emerging paradigm for achieving this goal. A VLA model integrates visual perception, language understanding, and action generation within a single neural network: given a camera observation and a natural language instruction, it directly outputs motor commands for a robot to execute. This end-to-end formulation eliminates the need for hand-crafted perception and planning pipelines, and allows the model to leverage the broad world knowledge encoded in its pre-trained VLM backbone.

Recent VLA systems have shown that this paradigm can already produce capable robots. π_0 [1] and $\pi_{0.5}$ [10] execute multi-step household tasks in previously unseen kitchens; GR00T N1 [25] extends this to humanoid platforms; DexVLA [31] targets bimanual dexterous manipulation (Figure 1). Crucially, recent evidence confirms that both ends of the VLA training pipeline are powerful scaling levers: pretraining on diverse data yields predictable performance gains [29], with human video data exhibiting log-linear scaling [36], and post-training through reinforcement learning from deployment experience can substantially reduce failure rates [11] (Section 2.1).

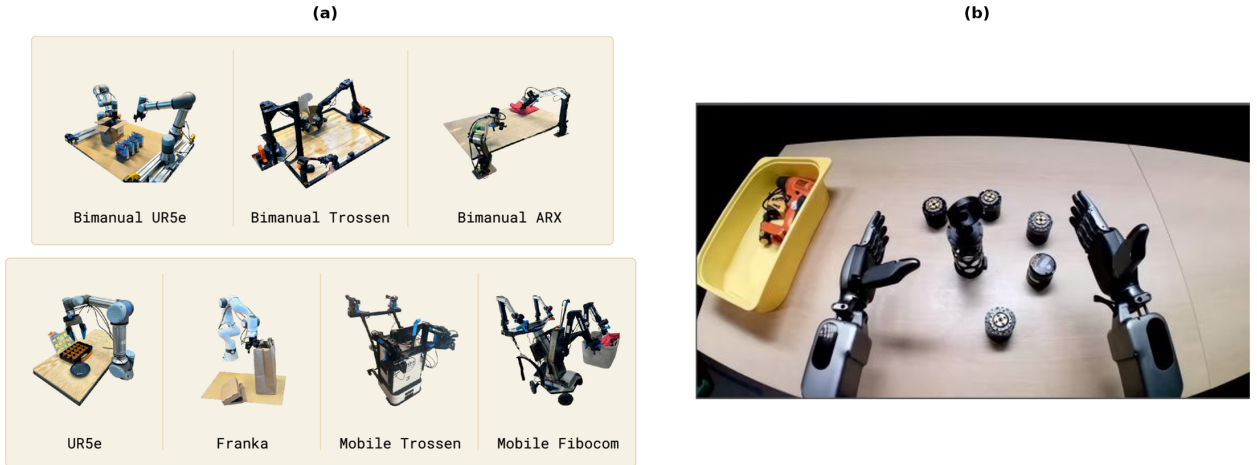


Figure 1: Representative VLA-enabled robotic systems. (a) Diverse robotic arm platforms used in large-scale VLA training [1]. (b) A humanoid robot with dexterous hands executing manipulation tasks conditioned on language instructions [25].

However, the VLA training pipeline faces two systemic bottlenecks that prevent these levers from scaling further (Figures 2 and 3). The first lies in pretraining: the data used to train VLA models suffers from severe modal incompleteness and annotation sparsity. The second lies in post-training: current methods for improving deployed policies remain heavily dependent on costly human intervention. These two bottlenecks are in fact two faces of the same underlying

(a) Pretraining: Modal Incompleteness

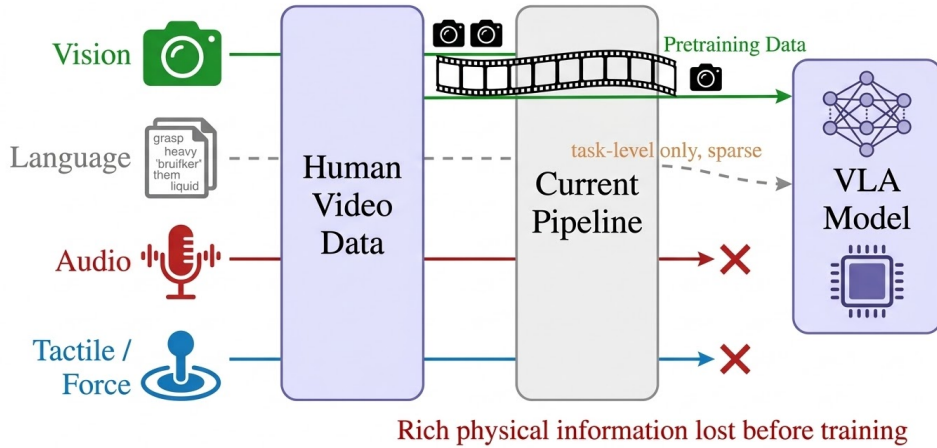


Figure 2: Pretraining bottleneck: modal incompleteness.

problem, namely that the VLA pipeline relies too heavily on scarce, human-generated resources, both curated multimodal data for pretraining and real-time human supervision for post-training. Addressing both is essential for scaling VLA systems to the breadth of tasks required for general-purpose robotic manipulation.

Bottleneck 1: modal incompleteness and sparse annotation in pretraining. Collecting robot demonstration data is prohibitively expensive, so recent work has turned to pre-training on large-scale human videos [21, 34, 36] as a scalable alternative (Section 2.2). However, these approaches systematically discard critical modalities. The audio channel, which carries rich information about contact events, material properties, and task progression, is stripped from the data before training begins. Furthermore, the language annotations in existing datasets remain coarse-grained, providing only task-level labels such as “tidy the desk” rather than frame-level descriptions of ongoing actions, which limits fine-grained language-action alignment. A key observation motivating this proposal is that both problems admit surprisingly simple solutions: audio is already present in human videos and needs only to be retained, and operator narration during recording can provide frame-level language annotations at near-zero additional cost.

Bottleneck 2: human dependency in post-training. Even if pretraining data were perfectly annotated, no static dataset can anticipate every situation a robot will face in the real world. Post-training, the process of improving a deployed policy from its own experience, is therefore indispensable. The current state of the art, RECAP [11], has demonstrated impressive gains (more than $2\times$ throughput, roughly halving failure rates), but it relies on two mechanisms that limit scalability (Section 2.3). First, every failed rollout requires a human operator to diagnose the failure and provide a correction via teleoperation, a cost that grows linearly with task diversity. Second, the reward signal is episode-level, with no semantically grounded intermediate feedback; for long-horizon tasks involving multiple manipulation steps, this makes credit assignment slow and inefficient.

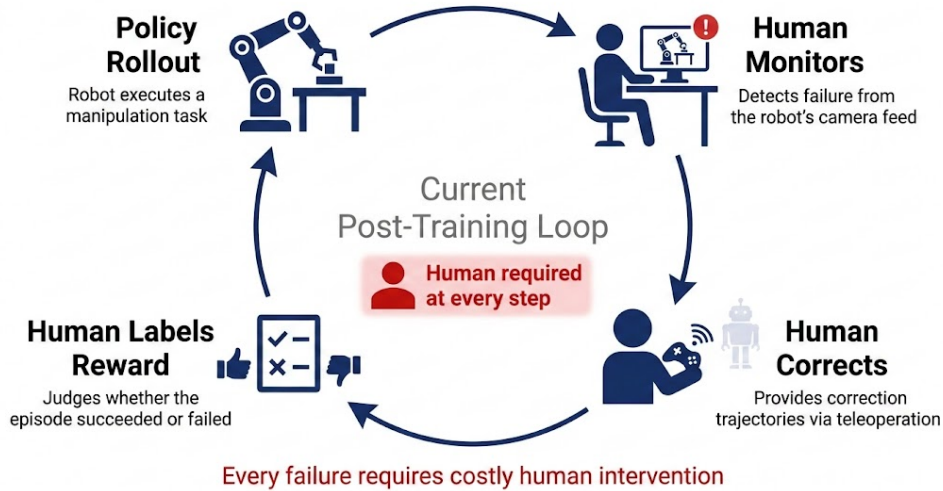


Figure 3: Post-training bottleneck: human dependency. The current post-training loop requires a human operator at three out of four stages: monitoring rollouts for failures, providing corrective trajectories via teleoperation, and labelling episode-level rewards. This cost scales linearly with task diversity.

Why solving these bottlenecks matters. Overcoming both bottlenecks would unlock a *closed-loop data flywheel* for robot learning: richer pretraining data produces better initial policies, autonomous post-training improves them from deployment experience without human intervention, and the improved trajectories flow back as higher-quality training data for the next cycle. Such a flywheel would allow VLA systems to scale rapidly to new tasks, new environments, and new robot embodiments, which is precisely the capability needed for practical applications in household service, industrial assembly, warehouse logistics, and assistive care, where task diversity is vast and human supervision is expensive. Recent work on foundation-model agents for reward generation [22, 30], task decomposition [35], and autonomous data collection [16] suggests that the building blocks for such a system are now within reach (Section 2.4), but no existing work unifies them into a coherent pipeline that spans both pretraining and post-training.

1.2 Research Objectives

The ultimate goal of this research is to build a **closed-loop, self-improving VLA training system** that eliminates human bottlenecks at both ends of the pipeline, enabling VLA models to scale autonomously to new tasks, environments, and embodiments. To achieve this, we propose a unified training paradigm with two complementary components: a **full-modality data pipeline** that enriches pretraining with audio signals and dense operator narration, and an **agent-in-the-loop post-training framework** that replaces human coaching with autonomous VLM/LLM agents and substitutes episode-level rewards with dense, frame-level evaluative feedback. Together, these components form a data flywheel: the pretraining pipeline produces capable initial policies, the agent-driven post-training loop improves them from deployment experience, and the improved trajectories feed back into pretraining, closing the loop with progressively decreasing human intervention. We decompose this goal into four sub-objectives:

1. Develop an audio-driven dense annotation pipeline that transforms human manipulation recordings into full-modality training data (vision, language, audio), with frame-level language descriptions derived from operator narration.
2. Design an agent-guided autonomous recovery mechanism that replaces human teleoperation coaching in the RECAP framework, using VLM analysis of failure states to generate semantic recovery instructions.
3. Construct an agent-based dense reward model that provides frame-level RL rewards, replacing episode-level success/failure signals with semantically grounded evaluative feedback from a VLM.
4. Integrate these components into a unified pretraining-to-post-training pipeline where trajectories from deployment flow back into training under a consistent semantic framework.

This work builds on our group’s prior research in two areas. DexSkills [23] demonstrated that haptic signals can segment manipulation demonstrations into reusable skill primitives; our pretraining pipeline extends this insight by using audio and narration for temporal segmentation of human video data. Our group’s work on dexterous arm-hand VLA [5] established shared autonomy approaches for bridging human intent and robot dexterity; the agent-in-the-loop post-training framework provides a scalable alternative to human shared control.

1.3 Structure Preview

The remainder of this proposal is organised as follows. Section 2 reviews related work across four areas: VLA models, data scaling and pretraining, post-training and RL, and agent-assisted robot learning. Section 3 presents our proposed methodology in detail, covering both the full-modality pretraining pipeline and the agent-in-the-loop post-training framework. Section 4 outlines the research timeline, publication plan, and expected milestones. Section 5 summarises the proposal.

2 Related Work and Open Challenges

2.1 Vision-Language-Action Models for Robot Manipulation

Vision-Language-Action (VLA) models unify visual perception, language understanding, and action generation within a single neural network. Since RT-2 first demonstrated that a VLM could be fine-tuned to output robot actions directly, the field has diversified along several architectural dimensions (Table 1).

Autoregressive models such as OpenVLA [14] tokenize continuous actions and predict them sequentially, inheriting the scaling properties of language models. OpenVLA achieves strong performance with 7B parameters on 970k real robot demonstrations. *Flow-matching models* such as π_0 [1] generate action chunks conditioned on visual and language features, handling multi-modal action distributions naturally. $\pi_{0.5}$ [10] extends this to open-world generalisation across household tasks by co-training on robot demonstrations, human video, and web-scale vision-language data. *Hybrid architectures* combine both paradigms: GR00T N1 [25] pairs a

Table 1: Comparison of representative VLA models. DiT = Diffusion Transformer. “—” indicates information not publicly available.

Model	Year	Params	Architecture	Data Source	Post-Train
OpenVLA	2024	7B	Autoregressive	Robot demo	Fine-tuning
π_0	2024	—	Flow matching	Robot demo + human video	—
$\pi_{0.5}$	2025	—	Flow matching	Robot demo + human video + web VL	—
$\pi_{0.6}$	2025	—	Flow matching	Robot demo + rollout + coaching	RECAP
GR00T N1	2025	—	Hybrid (VL + DiT)	Robot demo + human video + synthetic	—
DexVLA	2025	—	Hybrid (VLM + DiT)	Robot demo (cross-embodiment)	—
SmolVLA	2025	<1B	Autoregressive	Robot demo (community)	—
Evo-1	2025	0.77B	Hybrid (AR + DiT)	Downstream only	—
GR-3	2025	—	Hybrid (VLM + flow)	Robot demo + human video + web VL	Fine-tuning

vision-language module with a diffusion transformer for motor control; DexVLA [31] adds a plug-in diffusion expert for cross-embodiment dexterous manipulation; GR-3 [2] co-trains on web-scale vision-language data, VR-collected human trajectories, and robot demonstrations, outperforming π_0 on diverse tasks. Recent work has also pushed toward lightweight models: SmolVLA [28] maintains competitive performance at under 1B parameters, and Evo-1 [18] achieves state-of-the-art simulation results with only 0.77B parameters and no robot pretraining data.

Despite their impressive capabilities, VLA models share two structural limitations. First, they are data-hungry: even models that leverage VLM pre-training require thousands of hours of teleoperation data for reliable contact-rich manipulation. Second, while post-training methods are emerging (Section 2.3), they remain limited: current approaches either require human coaching for every failed rollout [8, 11] or rely on sparse, episode-level reward signals [15]. A practical concern is also inference latency: autoregressive models process tokens sequentially, leading to multi-hundred-millisecond inference times that can conflict with real-time control requirements at 10–50 Hz, while diffusion-based models require multiple denoising steps. Lightweight architectures such as Evo-1 and SmolVLA partially address this, but the tradeoff between model capacity and deployment speed remains an open challenge.

2.2 Data Scaling and Pretraining for VLA

A central question in VLA research is how to acquire sufficient training data. Two complementary scaling strategies have emerged: scaling real robot data across embodiments and tasks, and pre-training on large-scale human videos. A third emerging direction uses video world models to generate synthetic training experience.

2.2.1 Scaling Real Robot Data

Open X-Embodiment [4] aggregated demonstrations from 22 robot embodiments into a unified dataset, showing that cross-embodiment pre-training yields positive transfer. A Careful Examination of Large Behavior Models [29] rigorously validated this finding: multi-task pre-training makes policies more successful and robust, with performance increasing monotonically

as data scale and diversity grow. Crucially, the benefit is largest under distribution shift, where pre-trained models significantly outperform single-task baselines. A follow-up systematic study [17] investigated co-training strategies across five data modalities (vision-language data, dense language annotations, cross-embodiment robot data, human videos, and discrete action tokens) and found that co-training with vision-language and cross-embodiment data substantially improves generalisation, while training exclusively on robot data degrades the VLM backbone’s visiolinguistic understanding. These results establish that data scaling works, but also that *what* data is used matters as much as *how much*.

2.2.2 Pre-training on Human Videos

Large-scale egocentric video datasets such as Ego4D [7] (3,670 hours) and EgoDex [9] (829 hours with 3D hand tracking) provide abundant footage of humans performing manipulation tasks. Since these videos lack robot action labels, the dominant approach is latent action pre-training: an encoder learns to predict a latent vector that explains the visual transition between consecutive frames, which is later aligned with the robot’s action space. CLAP [34] uses contrastive learning to produce latent actions that are semantically consistent across human and robot domains. JALA [21] jointly aligns latent actions with robot joint spaces during pre-training on a 7.5M-clip video corpus spanning over 2,000 hours. At the data scaling end, EgoScale [36] pre-trains on 20,854 hours of egocentric human video and establishes a log-linear scaling law ($R^2 = 0.9983$) between data volume and downstream robot performance on 22-DoF dexterous hand tasks. An alternative line of work takes a human-centric view: Being-H0.5 [20] trains a Mixture-of-Flow architecture on 35,000 hours across 30 robot platforms to decouple shared motion primitives from embodiment-specific details, though cross-embodiment transfer for contact-rich dexterous tasks remains fragile. EgoVLA [32] focuses on egocentric pre-training with eye-gaze prediction, and Motion Tracks [27] extracts spatial trajectories from human demonstrations for few-shot transfer.

2.2.3 World Models for Data Generation

Rather than collecting more real data, an emerging direction trains video world models to generate synthetic training experience. DreamDojo [6] scales to 44,000 hours of human video to train a world model with latent actions that supports both teleoperation and model-based planning, effectively converting passive video into an interactive training environment. Cosmos Policy [13] fine-tunes a video foundation model for visuomotor control and planning, learning to predict future visual states conditioned on actions. These approaches complement direct data scaling by creating synthetic experience that can be generated at arbitrary volume. However, the fidelity of generated experience remains limited for contact-rich manipulation, where subtle physical interactions (friction, deformation, acoustic feedback) are difficult to simulate accurately in video space.

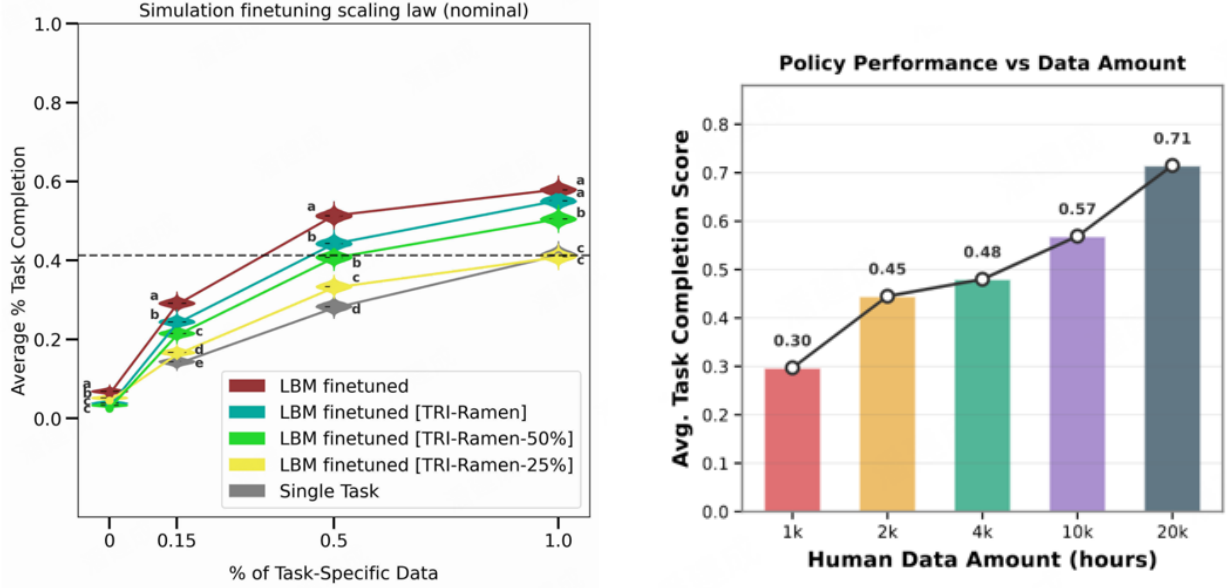


Figure 4: Scaling laws for VLA pretraining. Left: performance scales monotonically with real robot data diversity and volume [29]. Right: log-linear scaling between human video data volume and downstream dexterous manipulation performance [36].

2.2.4 Limitations: A Data Quality Ceiling

The scaling results above (Figure 4) confirm that more data helps, but a closer examination reveals that current pipelines hit a *data quality ceiling* well before data volume becomes the binding constraint.

Modal incompleteness. Every latent action method reviewed above (CLAP, JALA, EgoScale, Being-H0.5) strips audio from its input pipeline as a preprocessing step, treating human videos as silent films. This is a significant loss: manipulation audio encodes contact timing (when a grasp makes contact, when an object is placed down), material properties (metal vs. plastic vs. fabric), and interaction quality (a clean insertion click vs. a misalignment scrape). These signals are dense, temporally precise, and complementary to vision, which can be occluded by the hand during contact-critical moments. Retaining audio requires no additional data collection effort, only a change in the preprocessing pipeline.

Annotation sparsity. Existing datasets annotate at the task level (“make coffee”, “set table”) rather than the frame level (“reaching for mug”, “grasping handle”, “pouring water”). This coarse supervision means the model cannot learn fine-grained associations between specific visual transitions and their linguistic descriptions, limiting instruction-following precision. The contrast with the co-training study [17] is instructive: that work found that adding dense language annotations to robot trajectories improves generalisation, suggesting that annotation density, not just data volume, is a key driver of performance.

Data curation. While scaling laws show that aggregate data volume predicts performance, the optimal composition of heterogeneous sources (human video, robot demonstrations, synthetic rollouts, vision-language web data) remains poorly understood. The co-training study [17]

provides initial evidence that certain combinations are synergistic (vision-language co-training restores capabilities lost by robot-only training), but principled curation strategies that generalise across task domains are still lacking.

2.3 Post-Training and RL for VLA Policies

As noted in Section 1, post-training is essential for bridging the gap between pre-trained VLA capabilities and real-world deployment requirements. Here we examine the specific mechanisms that existing methods employ, their technical trade-offs, and why these trade-offs motivate our approach.

2.3.1 Human-in-the-Loop Approaches

$\pi_{0.6}$ [11] introduced RECAP (RL with Experience and Corrections via Advantage-conditioned Policies), which operates in two phases. In the first phase, the system collects deployment data: autonomous rollouts, human-coached corrections when rollouts fail, and expert demonstrations. The key technical insight is how RECAP handles this heterogeneous data: it trains a value function $V(s)$ that predicts the negative number of remaining steps until task completion, normalised to $(-1, 0)$, and uses N -step advantage estimates to label each trajectory segment by its expected improvement over the current policy. Trajectories with positive advantage (those that lead to faster completion than average) are upweighted during policy optimisation, while negative-advantage trajectories are downweighted. This advantage-conditioning mechanism allows RECAP to learn from both successful and failed episodes without requiring explicit reward engineering. In the second phase, the improved policy is redeployed, generating higher-quality data for the next training round. On tasks such as laundry folding in real homes and espresso preparation, this iterative process yields substantial cumulative gains.

The critical bottleneck, however, is the coaching step: a human operator monitors the rollout in real time and intervenes via teleoperation to correct mistakes as they occur. The authors note that even expert operators cannot guarantee consistent intervention quality, and that not all rollouts require coaching, but the operator must nonetheless be present and attentive throughout data collection to catch failures as they happen.

DexHiL [8] adapts this paradigm to dexterous manipulation, where the coaching burden is even greater. Multi-fingered hands have 16–22 degrees of freedom, and providing real-time corrections through teleoperation requires specialised interfaces (e.g., glove-based retargeting). DexHiL addresses data efficiency through an intervention-aware sampling strategy: rather than weighting all data equally, it identifies the segments immediately surrounding human interventions and over-samples these corrective segments during training, since they contain the most informative signal about failure recovery. This strategy improves success rates by 25% over standard offline fine-tuning, but does not reduce the fundamental requirement for human presence during data collection.

2.3.2 RL and Engineering Approaches

SimpleVLA-RL [15] explored scaling RL-based post-training across multiple tasks simultaneously. On LIBERO [19] and CALVIN [24], SimpleVLA-RL showed that multi-task RL can improve a pre-trained VLA without catastrophic forgetting of other tasks. The approach uses a simple online RL formulation with binary task-completion rewards. While the results demonstrate that RL post-training scales across task diversity, the reliance on binary rewards limits learning efficiency: the agent receives no feedback until it accidentally completes the full task, which becomes increasingly unlikely as task horizon grows.

χ_0 [33] addresses a different aspect of the post-training challenge: the distributional shift between human demonstrations, the policy’s learned behaviour, and the conditions encountered at deployment. χ_0 proposes three technical pillars. First, *Model Arithmetic*: instead of fine-tuning a single model, χ_0 trains separate models on different demonstration subsets (varying object appearances, initial states) and merges them in weight space, allowing the final policy to cover diverse conditions without requiring all variations in a single training run. Second, *Stage Advantage*: for multi-stage tasks, χ_0 defines stage-aware advantage estimates that provide dense progress signals at each stage boundary (e.g., “garment flattened”, “first fold completed”), rather than waiting for overall task success. This directly addresses the sparse-reward problem, and the authors show it produces more stable training than non-stage-aware baselines. Third, *Train-Deploy Alignment*: spatio-temporal augmentation, heuristic DAgger corrections, and temporal chunk smoothing bridge the gap between training and deployment distributions. Together, these components enable χ_0 to run dual-arm garment manipulation autonomously for 24 consecutive hours, surpassing $\pi_{0.5}$ by nearly 250% in success rate with only 20 hours of demonstration data.

2.3.3 Limitations and Open Questions

The methods reviewed above reveal a clear pattern: post-training works, but every existing approach makes trade-offs that limit scalability.

Coaching scalability. RECAP and DexHiL demonstrate that human corrections provide high-quality learning signal, but the requirement for real-time human presence during rollouts creates a linear cost that does not amortise across tasks. If a system must support 100 tasks, it needs 100 tasks’ worth of human monitoring time. No existing work has proposed a mechanism to generate comparable coaching signal autonomously.

Reward granularity. RECAP’s value function provides temporal signal (“how many steps remain”) but not semantic signal (“what went wrong”). χ_0 ’s Stage Advantage provides semantic progress tracking but requires manually pre-defined stage boundaries for each task, limiting generalisation to new tasks without additional human specification. SimpleVLA-RL’s binary reward provides no intermediate signal at all. An ideal reward system would provide frame-level semantic feedback without requiring per-task manual design.

Distribution shift. χ_0 demonstrates that addressing distributional inconsistency is critical for production-level robustness, but its solutions (Model Arithmetic, DAgger corrections) are task-

specific engineering. A general-purpose mechanism for adapting to deployment-time distribution shift remains an open problem.

2.4 Agent-Assisted Robot Learning

2.4.1 Agents in Other Domains

The rapid progress of foundation-model agents in other domains suggests a path toward automating the human-intensive components of the robot learning pipeline. In software engineering, agentic coding tools such as Claude Code [3] and Codex [12] autonomously read codebases, write and edit code, run tests, and iteratively fix errors, closing the loop between intent and execution without human intervention at each step. General-purpose agent frameworks such as OpenClaw [26] extend this autonomy to broader tasks, operating computers, browsers, and file systems through multi-step reasoning. The core capabilities that enable these systems, contextual understanding, multi-step planning, and closed-loop self-correction, are precisely the capabilities needed to replace human operators in the robot training pipeline.

2.4.2 Agents for Robot Learning

Within robotics, foundation-model agents have begun to address specific components of the training lifecycle. On the reward design front, Eureka [22] uses an LLM to generate reward functions from task descriptions, iteratively refining them based on training statistics; across 29 environments, Eureka-generated rewards match or exceed expert-designed ones. RL-VLM-F [30] takes a complementary approach, querying a VLM to express preferences between trajectory frames and training a reward model from these preferences, eliminating task-specific reward engineering entirely. For structured task decomposition, GRAPE [35] uses a VLM to decompose long-horizon tasks into temporal phases and generates preference-based cost functions for each phase, enabling more efficient learning on multi-step tasks. At the system level, RoboClaw [16] proposes an agentic framework for autonomous data collection, reducing human intervention by over 50%.

These works demonstrate that foundation-model agents can serve as reward designers, task decomposers, and data collectors. However, they address isolated components: Eureka and RL-VLM-F focus on reward generation, GRAPE on task decomposition, and RoboClaw on scene management. No existing work unifies these capabilities into a coherent agent-in-the-loop training paradigm that spans both pre-training and post-training. Furthermore, none of these methods address the data-side bottleneck of modal incompleteness and annotation sparsity, treating the pre-training data pipeline as given.

2.5 Summary of Research Gaps

Table 2 summarises eight open research gaps identified from the preceding review. We organise them into two groups: four gaps that this proposal directly addresses (marked **This work**), and four broader gaps that we acknowledge but do not target as primary contributions.

The four gaps targeted by this proposal form a coherent logic chain spanning both ends of the

Table 2: Research gap analysis. Gaps 1–4 are the focus of this proposal; Gaps 5–8 are open challenges acknowledged but not directly addressed.

#	Dimension	Current State	Proposed Direction	Ours
1	Modal completeness	Audio discarded by all latent action methods	Retain audio for dense contact event labels	Yes
2	Annotation density	Task-level language labels only	Operator narration for frame-level grounding	Yes
3	Human dependency	RECAP / DexHiL need human coaching per failure	VLM agent for failure analysis and recovery	Yes
4	Reward density	Episode-level success/failure only	Frame-level dense reward from VLM feedback	Yes
5	Cross-embodiment	Transfer fragile for contact-rich tasks	Embodiment-agnostic representations	No
6	Robustness	Distribution shift degrades performance; fixes are task-specific	General robustness and uncertainty quantification	No
7	Inference latency	Large VLAs too slow for 10–50 Hz control	Lightweight architectures and efficient inference	No
8	Data curation	Optimal composition of heterogeneous sources unclear	Principled curation for mixed-modality data	No

VLA pipeline. On the pre-training side (Gaps 1–2), incorporating audio and operator narration yields richer, more densely annotated training data, raising the data quality ceiling that currently limits scaling returns. On the post-training side (Gaps 3–4), replacing human coaching with an autonomous agent removes the scalability ceiling, while agent-generated dense rewards accelerate RL convergence on long-horizon tasks. These two directions are mutually reinforcing: higher-quality pre-training produces policies that require less post-training correction, while a more capable post-training loop can compensate for remaining gaps in pre-training coverage. Together, they form the closed-loop data flywheel described in Section 1.2.

3 Proposed Methodology

We propose a two-stage framework that addresses the four research gaps identified in Section 2.5. Stage 1 builds a full-modality pretraining pipeline that retains audio and adds operator narration to human video data, directly addressing Gaps 1–2 (modal incompleteness and annotation sparsity). Stage 2 introduces an agent-in-the-loop post-training framework in which a VLM agent replaces human coaching and provides dense reward signals, addressing Gaps 3–4 (human dependency and reward sparsity). Together, the two stages form a closed-loop data flywheel: deployment trajectories from Stage 2 flow back as training data for Stage 1, enabling continuous self-improvement with decreasing human involvement.

3.1 Problem Formulation

We assume access to two data sources. The first is a large-scale human video dataset $\mathcal{D}_h = \{(V_i, A_i)\}_{i=1}^{N_h}$, where V_i is a sequence of visual frames and A_i is the corresponding audio stream. Existing datasets such as Ego4D and EgoDex already contain audio but discard it during preprocessing; our pipeline retains A_i . For a subset of videos, we additionally collect operator narration N_i during recording, producing an enriched dataset $\mathcal{D}_h^+ = \{(V_i, A_i, N_i)\}$ with frame-

level language descriptions. The second data source is a smaller robot demonstration dataset $\mathcal{D}_r = \{(o_t, a_t, l)\}$ with observations o_t , actions a_t , and task-level language instructions l .

The goal is to learn a policy $\pi_\theta(a_t | o_t, l)$ that (1) benefits from pre-training on $\mathcal{D}_h$ by leveraging all available modalities, and (2) improves autonomously after deployment through agent-guided post-training, without requiring human intervention at each failure. We decompose this into two stages:

- **Stage 1** transforms $\mathcal{D}_h$ into full-modality annotated data and uses it to pre-train a VLA model.
- **Stage 2** deploys the pre-trained policy, uses a VLM agent to evaluate rollouts, generate dense rewards, and orchestrate recovery, then performs RL to improve π_θ .

3.2 Overall Framework

Figures 5 and 6 illustrate the two stages. The key design principle is that human involvement decreases progressively across the pipeline. In Stage 1, the only human contribution is narrating during video recording, a low-cost activity that produces high-value frame-level annotations. In Stage 2, the VLM agent progressively takes over the roles that RECAP assigns to human operators: failure diagnosis, corrective guidance, and reward labelling. As the system accumulates deployment experience, both the main policy and a learned recovery policy improve, further reducing the need for human fallback.

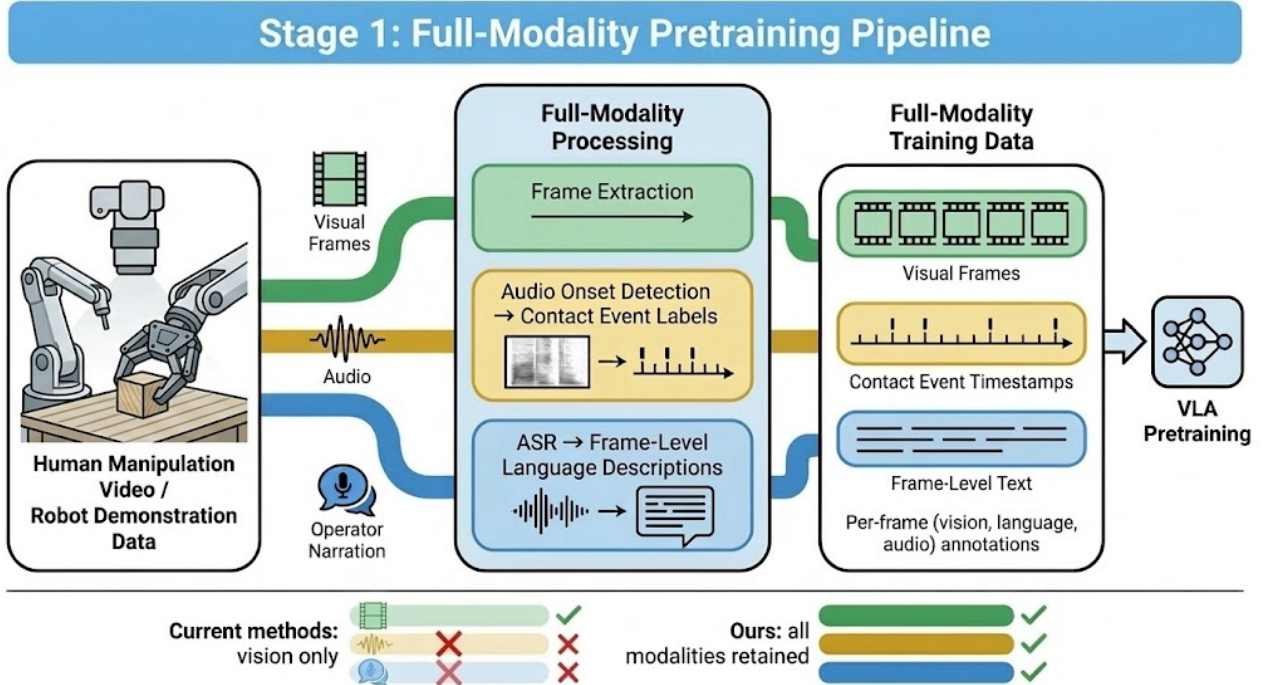


Figure 5: Stage 1: Full-modality pretraining pipeline. Human manipulation videos are processed to retain visual frames, audio contact events (via onset detection), and frame-level language descriptions (via ASR on operator narration). The resulting full-modality data is used to pre-train the VLA model.

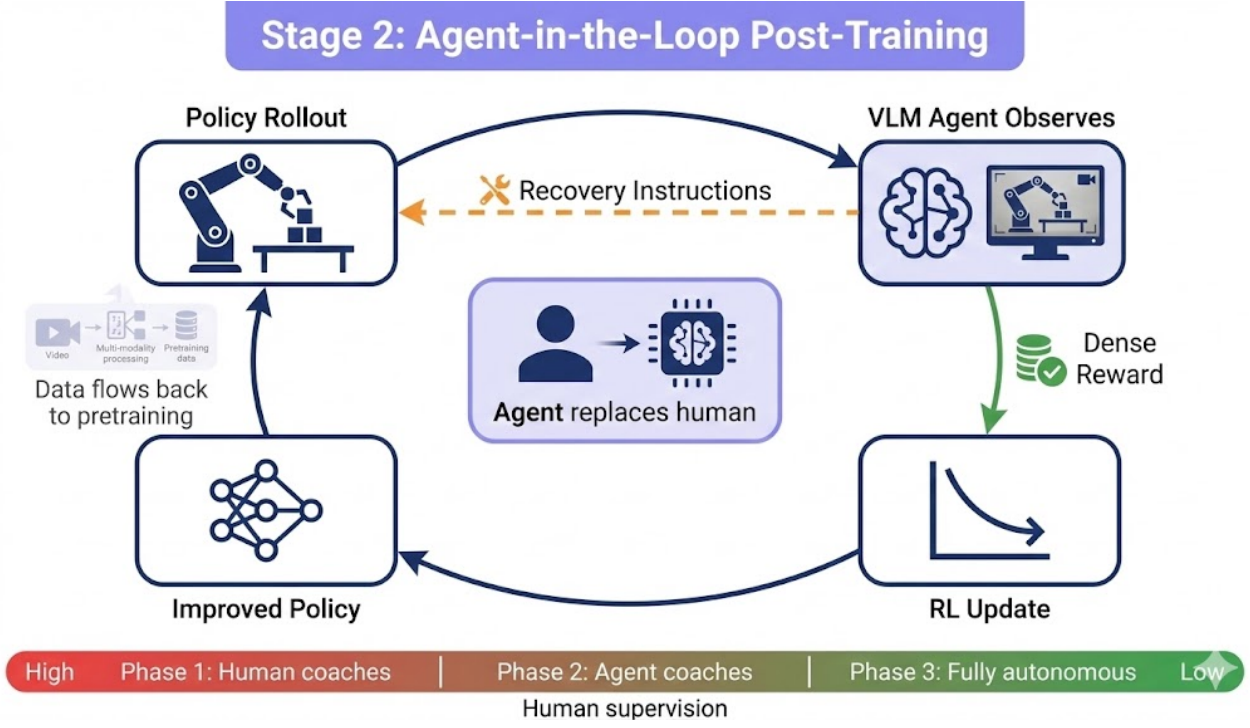


Figure 6: Stage 2: Agent-in-the-loop post-training. A VLM agent observes policy rollouts, detects failures, orchestrates recovery via a learned recovery policy, and generates frame-level dense rewards. Deployment trajectories flow back into the pretraining data pool, closing the self-improving flywheel.

3.3 Stage 1: Full-Modality Pretraining Pipeline

This stage transforms raw human manipulation videos into richly annotated training data by exploiting two modalities that current methods discard: audio and spoken narration.

3.3.1 Audio Event Detection and Temporal Annotation

Manipulation audio carries dense temporal information that is complementary to vision. Contact sounds mark the precise moments when a hand grasps an object, places it down, or inserts a component; acoustic signatures distinguish material interactions (metal, plastic, fabric); and the temporal pattern of audio events naturally segments continuous manipulation into discrete skill phases. Our group’s prior work DexSkills [23] demonstrated that haptic signals can segment demonstrations into reusable skill primitives; audio carries analogous contact information but can be captured passively from any video without requiring force sensors.

Our pipeline extracts the audio stream A_i from each human video and applies onset detection to identify contact events. Specifically, we compute the spectral flux of the audio signal and detect peaks that exceed an adaptive threshold, producing a sequence of timestamped contact events $\{(t_k, c_k)\}$ where t_k is the onset time and c_k is an event descriptor derived from the local spectral features. These contact events serve two purposes: (1) they provide dense temporal labels that mark when physically meaningful interactions occur, complementing the visual stream which can be occluded during contact; and (2) they provide candidate skill boundaries for temporal segmentation, since transitions between manipulation phases (reaching $\rightarrow$ grasping

→ transporting → placing) are typically accompanied by distinctive acoustic signatures.

3.3.2 Operator Narration and Dense Language Annotation

The second modality we exploit is operator narration. During video recording, we ask operators to narrate their actions in real time (e.g., “now reaching for the red cup”, “grasping the handle”, “pouring water into the bowl”). This narration stream N_i is processed through an automatic speech recognition (ASR) system to produce timestamped text transcriptions aligned with the visual frames. The result is a frame-level language description for the entire video, far denser than the task-level labels (e.g., “make coffee”) available in existing datasets.

This approach has three advantages over alternative annotation strategies. First, it is nearly zero-cost: narration is a natural byproduct of the recording process and requires no post-hoc labelling effort. Second, it captures the operator’s intent and attention at each moment, providing semantically grounded descriptions that reflect what the operator considers important. Third, it naturally varies in granularity with task complexity: simple actions receive brief descriptions while complex manipulations receive detailed step-by-step commentary.

The co-training study [17] found that adding dense language annotations to robot trajectories improves VLA generalisation, providing empirical support in the robot data setting for the value of annotation density. Operator narration extends this principle to human video data, offering a scalable way to produce the kind of dense language supervision that benefits downstream performance.

3.3.3 Full-Modality Data Integration

For each frame t in a human video, the pipeline produces a full-modality annotation tuple:

$$d_t = (I_t, \ell_t, e_t) \tag{1}$$

where I_t is the visual frame, ℓ_t is the ASR-derived language description active at time t , and $e_t \in \{0, 1\}$ indicates whether a contact event was detected at time t . This tuple replaces the vision-only frame used by all existing latent action methods.

The full-modality data is integrated into VLA pre-training through multi-modal conditioning: the visual encoder processes I_t , the language encoder processes ℓ_t (providing frame-level language grounding rather than task-level only), and the audio event label e_t provides an additional conditioning signal that marks contact-critical frames. The key point is that this requires minimal architectural modification to existing VLA pipelines; the primary change is in the data preprocessing, not the model architecture.

3.4 Stage 2: Agent-in-the-Loop Post-Training

Once the VLA policy is pre-trained and deployed, it inevitably encounters situations where its behaviour is insufficient. Stage 2 uses a VLM agent to automate the three functions that currently require human operators: failure detection, corrective intervention, and reward generation.

3.4.1 Agent-Guided Failure Recovery

In the RECAP framework, a human operator watches each rollout, detects failures in real time, and provides corrections via teleoperation. We replace this with a two-component system: a VLM agent for failure analysis and a learned recovery policy for physical correction.

The *VLM agent* observes the robot’s camera feed and continuously evaluates rollout progress. When it detects a failure (e.g., a dropped object, a missed grasp, a collision), it performs two actions. First, it generates a semantic failure diagnosis: a natural language description of what went wrong and why (e.g., “the grasp failed because the fingers closed before reaching the object”). Second, it selects and invokes an appropriate recovery strategy, either issuing language-based instructions to the main policy or activating a dedicated *recovery policy*.

The *recovery policy* π_{rec} is a separate policy trained specifically for failure recovery and scene restoration. It handles tasks such as releasing a misgripped object, returning the arm to a safe configuration, or resetting the workspace to a state from which the main policy can retry. Critically, the recovery policy is itself part of the self-improving loop: each recovery episode generates trajectory data that is added to the recovery training set, so π_{rec} improves as the system accumulates deployment experience. This creates a virtuous cycle where both the main policy (fewer failures over time) and the recovery policy (better recovery when failures do occur) improve in tandem.

The transition from human-dependent to fully autonomous post-training follows a phased approach:

1. **Phase 1** (human + agent): The VLM agent handles failure detection and reward generation; a human operator provides teleoperation corrections as fallback.
2. **Phase 2** (agent + recovery policy): The trained recovery policy handles physical corrections; the human monitors but rarely intervenes.
3. **Phase 3** (fully autonomous): Both failure recovery and reward generation are handled entirely by the agent and recovery policy; human involvement is limited to periodic quality audits.

3.4.2 Agent-Based Dense Reward Generation

Current post-training methods rely on episode-level reward signals that provide no semantic intermediate feedback (Section 2.3). We use the VLM agent to generate frame-level dense rewards that capture both task progress and execution quality.

We consider two complementary reward generation mechanisms. The first is *preference-based reward*, following the approach of RL-VLM-F [30]: the VLM is presented with pairs of trajectory segments and asked to express a preference for the segment that represents better task progress. A reward model is trained from these preferences, providing a continuous reward signal that captures the VLM’s notion of improvement without requiring explicit reward function design. The second is *direct VQA scoring*: at each frame, the VLM is queried with structured questions

(e.g., “Is the robot making progress toward the goal?”, “Is the grasp stable?”, “Has a sub-task been completed?”) and its responses are converted into a scalar reward. This approach is more interpretable and allows task-decomposed reward signals, building on the temporal phase decomposition of GRAPE [35].

Unlike χ_0 ’s Stage Advantage, which requires manually defined stage boundaries for each task, our agent-based reward is generated from the VLM’s general visual understanding and requires no per-task specification. The VLM can identify meaningful sub-goals (“object grasped”, “object transported to target zone”, “object placed upright”) directly from the visual context, generalising across tasks without additional human input.

3.4.3 RL Training Loop

The dense reward signal from the VLM agent is combined with a sparse task-completion signal to form the total reward:

$$r_t = r_{\text{task}} + \alpha \cdot r_{\text{agent}}(t) \quad (2)$$

where r_{task} is the binary task-success reward at episode termination, $r_{\text{agent}}(t)$ is the frame-level agent-generated reward, and α controls the balance between the two signals.

We adopt the advantage-conditioned policy optimisation framework from RECAP [11], extending it with our dense reward signal. The value function is trained to predict expected returns under the combined reward, providing advantage estimates that are informed by both task progress (from the agent) and overall success (from the task signal). This preserves RECAP’s ability to learn from heterogeneous data (demonstrations, on-policy rollouts, and recovery episodes) while providing much richer learning signal for credit assignment in long-horizon tasks.

The flywheel closes when deployment trajectories, now annotated with agent-generated rewards and failure analyses, flow back into the Stage 1 data pool. Successful trajectories augment the pretraining data; failure trajectories with recovery episodes train the recovery policy; and the VLM agent’s failure diagnoses provide a growing corpus of failure modes that can inform future reward design. This continuous loop is the core mechanism that enables scaling to new tasks and environments without proportional increases in human supervision.

3.5 Scope and Expected Contributions

This PhD focuses on the two-stage framework described above, with the following scope decisions:

- We target **tabletop dexterous manipulation** tasks (grasping, insertion, pouring, tool use) rather than mobile manipulation or locomotion.
- We use publicly available egocentric video datasets (Ego4D [7], EgoDex [9]) for Stage 1, supplemented by a small set of narrated recordings collected in our lab.
- We evaluate on established benchmarks (LIBERO [19], CALVIN [24]) for simulation and on our group’s dexterous arm-hand platform [5] for real-robot experiments.
- We do **not** address cross-embodiment transfer (Gap 5), deployment robustness beyond what the agent provides (Gap 6), inference latency optimisation (Gap 7), or principled

data curation strategies (Gap 8), though our framework is compatible with future work in these directions.

The expected contributions are:

1. A **full-modality pretraining pipeline** that retains audio and adds operator narration to human video data, producing frame-level multimodal annotations at near-zero additional cost.
2. An **agent-guided failure recovery** mechanism that replaces human teleoperation coaching with VLM-based failure analysis and a learned recovery policy, enabling autonomous post-training.
3. An **agent-based dense reward model** that provides frame-level RL rewards from VLM feedback, eliminating the need for per-task reward engineering or manually defined stage boundaries.
4. A **unified pretraining-to-post-training pipeline** that forms a closed-loop data flywheel, where deployment experience flows back into training under a consistent framework.

3.6 Challenges and Mitigation Strategies

Audio quality in egocentric videos. Datasets such as Ego4D were not recorded with manipulation audio in mind. Many clips contain background music, speech from bystanders, or low-fidelity audio. We will apply spectral gating for broadband noise reduction and train a lightweight audio quality scorer on a small curated subset to rank clips by manipulation-audio salience. We expect that focusing on the high-quality subset for initial development, then gradually expanding to noisier data, will yield a robust pipeline.

VLM reward reliability. VLMs can hallucinate or produce inconsistent evaluations. We mitigate this through three mechanisms: (1) calibrating agent rewards against a small set of human-labelled trajectories to detect systematic biases; (2) using ensemble voting across multiple VLM queries with rephrased prompts to improve consistency; and (3) treating agent rewards as a supplement to, not a replacement for, the sparse task-completion signal, so that hallucinated rewards cannot override ground-truth success/failure.

Recovery policy quality. Early in training, the recovery policy will have limited capability, potentially making situations worse rather than better. We address this through the phased transition described in Section 3.4: in Phase 1, human operators provide corrections while the recovery policy learns from observation; in Phase 2, the recovery policy handles simple recoveries while humans handle complex ones; in Phase 3, the recovery policy operates independently. This gradual handoff ensures that the system never relies on a recovery policy that has not been validated on sufficient data.

RL sample efficiency on real hardware. Real-robot rollouts are slow and expensive. We leverage three strategies: (1) offline RL with advantage conditioning (following RECAP), which reuses deployment data without requiring fully on-policy collection; (2) agent-generated dense

rewards that accelerate credit assignment; and (3) simulation-first validation on LIBERO and CALVIN before transferring to real hardware, reducing the number of real-robot iterations needed.

4 Research Plan

The expected contributions of this research are: (i) a full-modality pretraining pipeline that retains audio and adds operator narration to human video data; (ii) an agent-in-the-loop post-training framework with VLM-based failure recovery and dense reward generation; and (iii) a unified closed-loop pipeline connecting both stages into a self-improving flywheel. Detailed plans for each year are given below, with Gantt charts in Figures 7–9.

4.1 Phase 1: Full-Modality Pretraining Pipeline (Year 1, 2026–2027)

We will build the full-modality data pipeline (Section 3.3): extracting audio from Ego4D [7] and EgoDex [9], implementing onset detection for contact events, collecting narrated recordings in our lab, and setting up ASR for frame-level language annotations. In parallel, we will reproduce CLAP and JALA as baselines. In the second half, we will pre-train a VLA on the full-modality data and run ablations on LIBERO [19] and CALVIN [24] to evaluate the contribution of each modality. We target a workshop or conference paper at CoRL/RSS 2027.

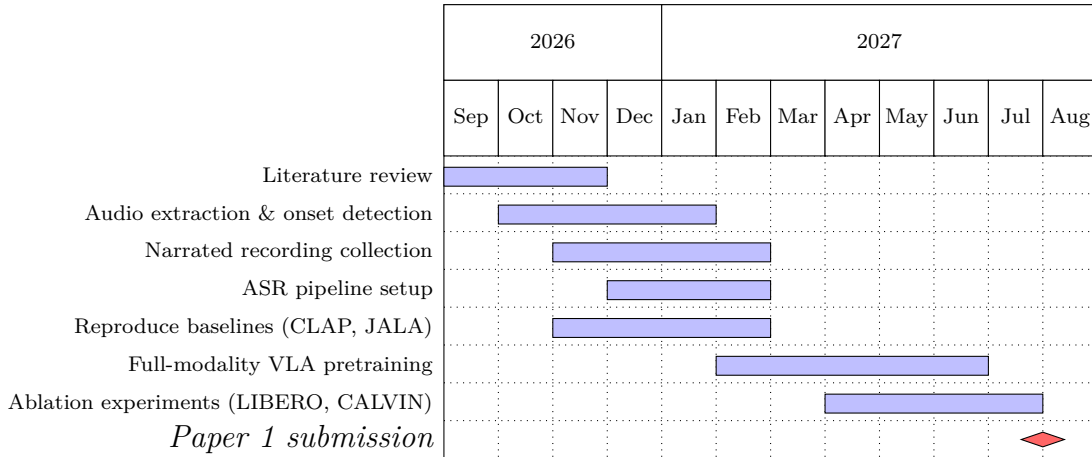


Figure 7: Year 1 plan: Full-modality pretraining pipeline.

4.2 Phase 2: Agent-in-the-Loop Post-Training (Year 2, 2027–2028)

We will implement the agent-in-the-loop post-training framework (Section 3.4): deploying the pre-trained VLA on simulation benchmarks, building the VLM agent for failure detection and dense reward generation, and training the initial recovery policy from human corrections (Phase 1 of the transition). In the second half, we will transfer to our group’s dexterous arm-hand platform [5], transition to Phase 2 (agent + recovery policy), and evaluate on contact-rich tasks. We target a full paper at CoRL/ICRA 2028.

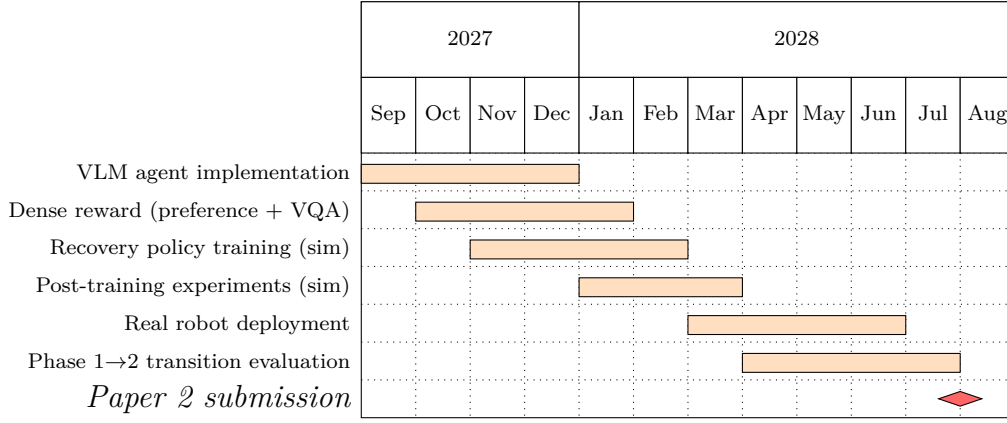


Figure 8: Year 2 plan: Agent-in-the-loop post-training.

4.3 Phase 3: Integration, Scaling, and Thesis (Years 3–4, 2028–2030)

In Year 3, we will close the flywheel by integrating Stages 1 and 2, evaluating whether multiple cycles of the loop produce measurable improvement, and scaling to long-horizon tasks (5+ steps). We target a conference paper at RSS/CoRL 2029. In Year 4, we will complete final ablations and allocate six months for thesis writing, following Prof. Li’s recommendation.

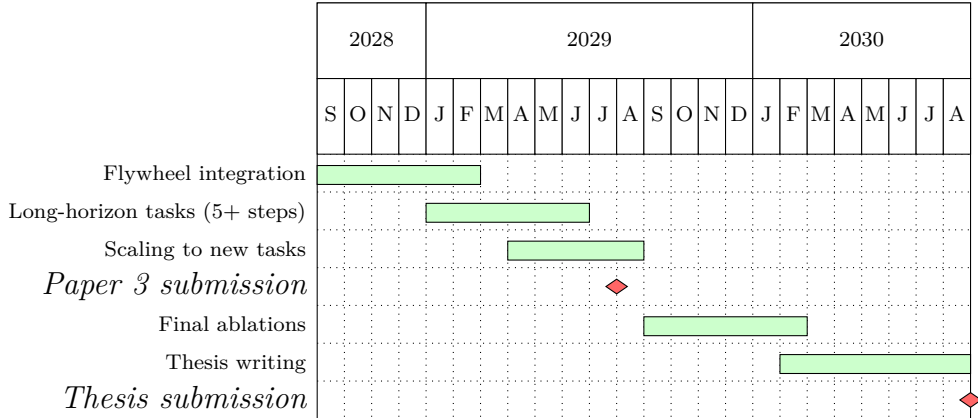


Figure 9: Years 3–4 plan: Integration, scaling, and thesis writing.

5 Summary

This proposal presents a two-stage framework for building self-improving Vision-Language-Action models that scale with progressively decreasing human involvement.

On the pretraining side (Stage 1), we address the data quality ceiling that limits current VLA pipelines by retaining the audio modality and adding operator narration to human video data. Audio onset detection provides dense temporal labels for contact events, while ASR-transcribed narration provides frame-level language descriptions, together producing full-modality annotations at near-zero additional cost. This directly addresses the modal incompleteness and annotation sparsity that all existing latent action methods suffer from.

On the post-training side (Stage 2), we replace the human operators that current methods

require with a VLM agent that autonomously detects failures, orchestrates recovery via a learned recovery policy, and generates frame-level dense rewards. This eliminates the scalability ceiling of human-in-the-loop coaching while providing semantically richer learning signals than episode-level rewards. Both the main policy and the recovery policy improve through the training loop, creating a virtuous cycle.

The two stages are connected by a closed-loop data flywheel: deployment trajectories from Stage 2 flow back as training data for Stage 1, enabling continuous improvement without proportional increases in human supervision. This flywheel is the core mechanism that enables scaling to new tasks, environments, and embodiments.

If successful, this research would demonstrate that the two dominant bottlenecks in VLA development, scarce multimodal pretraining data and human-dependent post-training, can be addressed within a single unified framework. The resulting system would allow VLA models to improve autonomously from deployment experience, reducing the marginal cost of supporting each new task and bringing general-purpose robotic manipulation closer to practical deployment in household, industrial, and assistive settings.

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